

Cooking Mama Mobile

Final Project CS388 Report

1. Game Description & Setup

Cooking Mama Mobile is a mobile casual game developed in Unity where the player takes on the role of a chef slicing ingredients. Fruits and vegetables move along a rail towards a stationary knife, and the player must tap and hold the screen to bring the knife down and slice them before they pass. To add some challenge, bombs are occasionally mixed in with the ingredients — accidentally cutting one will stun the knife temporarily. Two levels are included, each with its own sequence of ingredients defined via ScriptableObjects.

How to play:

- Tap and hold the screen to swing the knife down and slice incoming ingredients.
- Every ingredient successfully cut adds points to the score.
- Cutting a bomb stuns the knife — shake the phone hard to recover.
- Reach the Trophy at the end of the sequence to complete the level and unlock the next one.

2. Interaction Techniques

Touch Input

The main interaction is built around touch input. When the player taps and holds the screen, the knife moves downward to perform a cut. To make this feel more natural on mobile, a small time window was implemented so that the cutting state stays active for a short period after the touch is detected. This means the player does not need perfect timing — as long as their finger is on the screen close to when an ingredient arrives, the cut will register correctly.

Shake Gesture

A secondary interaction uses the phone's accelerometer. When the knife gets stunned by a bomb, the player needs to physically shake the device to de-stun it. Rather than unlocking instantly on a single jolt, the game accumulates the force of each shake over time until a target value is reached, so the player has to keep shaking for a moment.

3. Rendering & Asset Optimization

GPU Instancing

GPU instancing was enabled on all the fruit and vegetable slice materials. Since the same meshes are spawned repeatedly throughout a level, this allows Unity to group their draw calls together and send them to the GPU in one batch instead of individually, which helps keep performance stable on mobile hardware.

Mipmaps

All ingredient textures have mipmaps enabled in their import settings. Unity uses this to automatically swap in lower resolution versions of a texture when the object is far from the camera, which reduces memory usage and prevents visual artifacts.

Static Batching

Scene objects that never move, such as the cutting board and background elements, were marked as Static in the Unity Inspector. This lets Unity merge their geometry into a single draw call at build time, reducing the CPU work needed each frame to render the scene.

4. Research Topic: Sensor Fusion

The shake-to-de-stun mechanic in Cooking Mama Mobile is a practical example of sensor fusion. Rather than using the accelerometer data on its own, the game combines it with the current state of the knife (specifically whether it is stunned) to decide whether the sensor input should have any effect. Raw accelerometer readings are sampled every frame, filtered by a minimum magnitude threshold to ignore small unintentional movements, and accumulated over time. This fused signal — sensor data plus game state — then triggers the de-stun only when both conditions are met. This is consistent with the definition of sensor fusion as the process of combining data from multiple sources to produce information that would not be possible from any single source alone.

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